Objects

Exercise

Storing Values in Objects

Take a look at the code below:

```r
# Assign to an object called `x`, the result of `3 + 5`
x <- 3 + 5

# Assign to an object called `x`, the result of `x - 4`
x <- x - 4

# Assign to an object called `y`, the result of `x + 1`
y <- x + 1

# Print the value stored inside of `x`
print(x)

# Print the value stored inside `y`
print(y)
```

What are the values of x and y after this code runs?

a. x = 5, y = 5
b. x = 5, y = 4
c. x = 4, y = 5
d. x = 4, y = 4

Exercise

Storing Values in Objects

Take a look at the code below:

```
# Assign to an object called `x`, the result of `3 + 5`
x <- 3 + 5

# Assign to an object called `x`, the result of `x - 4`
x <- x - 4

# Assign to an object called `y`, the result of `x + 1`
y <- x + 1

# Print the value stored inside of `x`
print(x)

# Print the value stored inside `y`
print(y)
```

What are the values of x and y after this code runs?

a. x = 5, y = 5
b. x = 5, y = 4
c. **x = 4, y = 5**
d. x = 4, y = 4

Pro Tip: *The lines of code that start with # are called comments. Computers will ignore comments when running your code.*

What are Objects?

Objects (also called **variables**) store values for later use.

```
# Assign to an object called `x`, the result of `3 + 5`
x <- 3 + 5

# Assign to an object called `x`, the result of `x - 4`
x <- x - 4

# Assign to an object called `y`, the result of `x + 1`
y <- x + 1

# Print the value stored inside of `x`
print(x)

# Print the value stored inside `y`
print(y)
```

- In the exercise, x and y were objects.
- We used the **assignment operator** (`<-`) to assign values to x and y.

Rules and Recommendations for Object Names

Rules

- No spaces (e.g. `num obj`) or mathematical operators (`num-obj`).
- Can't start with underscores or numbers (e.g. `_obj` and `2obj`).
- Can include upper/lower case letters, numbers, underscores, and periods.
- Object names are case sensitive (e.g. `num_obj` and `Num_obj` are not the same).

Recommendations

- Use **snake_case**:
  - All lower case letters
  - use underscore(`_`) to separate words
- Use nouns that describe the value that the object contains
  - Avoid using verbs as object names (except for function names).

Knowledge Check

Object Names

Select all that apply:

Which of the following names follow our object name rules?

a. `blood_pressure`
b. `bp`
c. `bloodPressure`
d. `measure-BP`

Select all that apply:

Which of the following names follow our object name recommendations?

a. `blood_pressure`
b. `bp`
c. `bloodPressure`
d. `measure-BP`

Knowledge Check

Object Names

Select all that apply:

Which of the following names follow our object name rules?

a. `blood_pressure`
b. `bp`
c. `bloodPressure`
d. `measure-BP`

Select all that apply:

Which of the following names follow our object name recommendations?

a. `blood_pressure`
b. `bp`
c. `bloodPressure`
d. `measure-BP`

UCSF

University of California
San Francisco